Bocciolone Camilla
10783793
Stoppani Riccardo
10719433
Molino Salvatore
10794839
Monaco Pierpaolo
10804734
Cao Luisa 10835823

POLITECNICO
MILANO 1863

SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE

Segmenting Bank Clients: A Dual-Approach Comparison

1. Introduction

Customer segmentation is a **key task in banking**, as it enables institutions to **better understand their clients**, **personalize services**, **improve marketing strategies** and **support risk management decisions**. Depending on the bank's objectives, segmentation may require either a **coarse grouping of clients** into a few broad profiles or a **more granular partition** into many fine-grained personas. For this reason, we adress the problem following **two complementary approaches**: the main one is implemented in the **first notebook (NB1)** and is a more general segmentation while in **(NB2)** there is a more granular clusterization.
In this context, we consider a dataset that mixes **twelve numerical variables** (Age, Income, Wealth, Debt, FinEdu, ESG, Digital, BankFriend, LifeStyle, Luxury, Saving, FamilySize) with **five categorical ones** (Gender, Job, Area, CitySize, Investments).

2. Data Preparation common to NB1 and NB2

We started by preparing the dataset using an **80/20 train–test split**. Throughout the analysis, the training set was used to perform clustering and model development, while the test set was reserved for validation purposes, allowing us to assess the robustness and generalisability of the segmentation results.
After the split, we applied a **scaling procedure** to all numerical variables in order to place them on a comparable scale and avoid dominance effects in distance-based methods, while categorical variables were **one-hot encoded** using a drop-first strategy, thus creating a fully numerical input.

3. NB1

3.1. Distances and dimensional reduction

Since the dataset contained both numerical and categorical features, standard distances such as the Euclidean distance could not be directly applied. We therefore considered the *Gower distance*, which is suitable for mixed-type data and allows feature weighting. After testing alternative mixed-distance specifications, the **weighted Gower distance** was selected as it produced the most stable and interpretable clustering results. Higher weights were assigned to financial variables (*Income, Wealth, Debt, Saving*), moderate weights to behavioural variables (*Digital Usage, Financial Education, LifeStyle*) and lower weights to demographic variables (*Gender, Area, CitySize*).
To explore the potential cluster structure, we applied several **dimensionality reduction techniques**. Comparing embeddings from methods capturing different data properties (e.g. global variance in PCA and local neighbourhood structure in t-SNE) provided a more robust understanding of the segmentation and the number of clusters. These visualisations were used only as **exploratory support**.

3.2. Clustering and Model Selection

Since the true number and structure of clusters are unknown, we adopted a **multi-method clustering strategy**. Different algorithms capture different data structures (e.g. hierarchical organisation, density separation or probabilistic overlap). By comparing several paradigms on the same distance configuration, we aimed to obtain **robust and interpretable segmentations**. We therefore applied hierarchical clustering (average and complete linkage), K-Medoids, DBSCAN, Gaussian Mixture Models, Spectral Clustering, K-Prototypes and Self-Organising Maps, each selected for complementary strengths in handling mixed data, non-Euclidean distances or non-convex cluster shapes.
As clustering is an **unsupervised task**, no ground-truth labels were available. We therefore relied on **internal validation metrics** measuring cluster compactness and separation. In particular, we computed the Silhouette index, the Calinski–Harabasz index and the Davies–Bouldin index, which provide complementary geometric perspectives on clustering quality. To reduce sensitivity to individual metric biases, the optimal number of clusters was selected through a **median voting scheme** across the three indices. Finally, we assessed the **stability of our solutions** via bootstrap resampling and Jaccard similarity. This allowed us to verify whether the identified segmentation represented a reproducible structure rather than noise or sampling variability.
Overall, **hierarchical clustering with complete linkage and** $k = 4$ was selected as the best trade-off between segmentation granularity, internal validation performance (**Silhoutte of 0.210, CD of 680 and DB of 1.6**) and interpretability. This configuration yielded a sufficiently detailed yet stable customer partition, while the t-SNE embedding showed clearly separated and well-defined cluster regions.

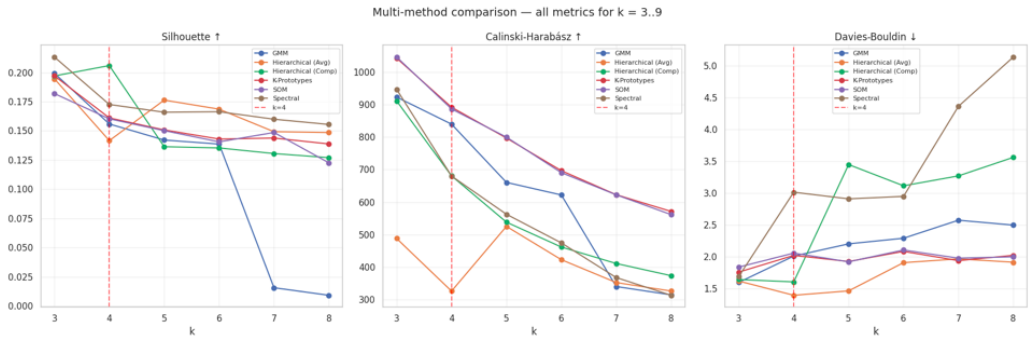


Figure 1: comparison of clustering performance across different methods and numbers of clusters

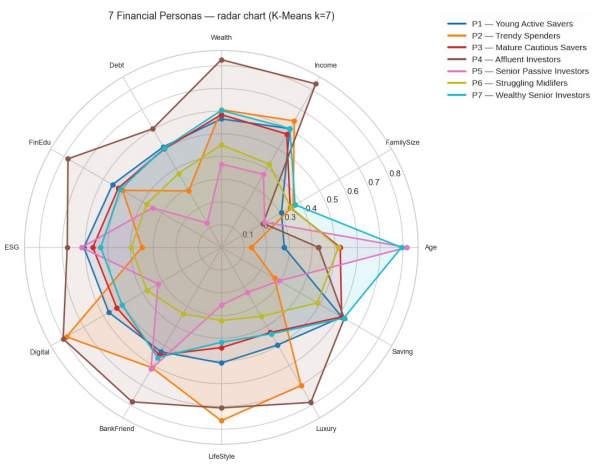
4. NB2

The second approach was built entirely on **K-Means clustering**, operating on the twelve numerical variables plus *Investments*. K-Means was run across $k \in \{3, \ldots, 11\}$ with `n_init=20`. The optimal k was identified by jointly evaluating the Silhouette index, the Calinski–Harabász index, the Davies–Bouldin index and the inertia elbow. $k = 7$ was selected as it occupied a consistently strong region across all metrics: silhouette improved slightly from $k = 6$, Davies–Bouldin dropped noticeably indicating better intra-cluster compactness and the inertia curve showed a clear elbow. Importantly, $k = 7$ was not the global optimum on any single metric, but it represented the best trade-off between cluster quality and interpretability.

5. Financial Personas

Table 1: NB1 — Four Personas

Persona	Key Traits
P1 — Elderly Conservative Low-Wealth	Avg. age 80, low income and wealth, very low debt. They show limited digital adoption, modest financial education and a simple lifestyle with low luxury spending.
P2 — Affluent Financially Engaged	Avg. age 50, high income and the highest wealth, but also high debt, reflecting active financial leverage. They are digitally savvy, financially educated and consumption-oriented, with strong saving capacity.
P3 — Young Digital Lifestyle	Avg. age 25, moderate income and wealth but low savings. They are highly digital and lifestyle-oriented, with moderate financial education and ESG awareness.
P4 — Mature Balanced Planners	Avg. age 59, moderate income, wealth and debt, showing consistent saving behaviour. They display balanced financial education, average digital usage and pragmatic consumption patterns.



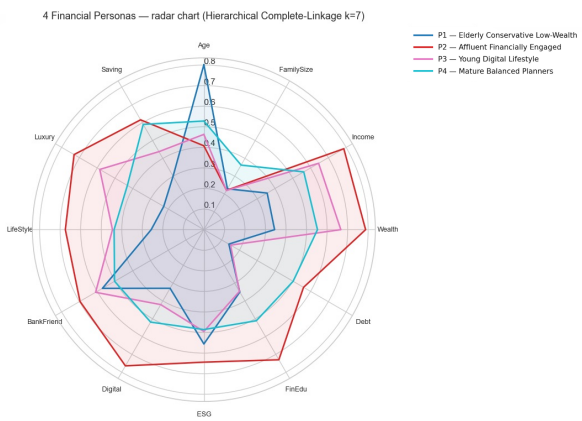


Table 2: NB2 — Seven Personas

Persona	Key Traits
P1 — Young Active Savers	Avg. age 40, young adults in their prime working years with a healthy income and a strong focus on saving. Digitally comfortable, with good financial habits and a moderate but active approach to investments. Lifestyle is balanced, without a strong emphasis on luxury spending.
P2 — Trendy Spenders	Avg. age 29, the youngest cohort, lifestyle-oriented and focused on high-end living. Very low savings and minimal investments, but highly digital and socially engaged with banking.
P3 — Mature Cautious Savers	Avg. age 59, middle-aged to early-retirement individuals with solid savings but a very conservative approach to investments. Modest lifestyle, prefer traditional banking, and prioritize financial safety.
P4 — Affluent Investors	Avg. age 52, high-net-worth clients with strong incomes and substantial wealth. Sophisticated investors with excellent financial knowledge, high digital affinity, and a premium lifestyle.
P5 — Senior Passive Investors	Avg. age 81, older retirees with modest but stable resources. Maintain a basic lifestyle, low debt, and moderate investments. Minimal digital engagement, rely on established banking relationships.
P6 — Struggling Midlifers	Avg. age 58, midlife segment facing financial challenges. Limited income and wealth, some debt, low financial literacy, and suboptimal investments. Low digital engagement, constrained lifestyles.
P7 — Wealthy Senior Investors	Avg. age 79, well-off retirees with substantial wealth and income. Balanced finances with moderate debt, high savings, and diversified investments. Digital engagement is average.

6. Test Set Generalisation

To assess the operational usability of the proposed segmentation, we evaluated how the identified personas generalised to unseen data using two complementary deployment strategies implemented in Notebook 1 (NB1) and Notebook 2 (NB2). In **NB1**, we trained a Generalised Learning Vector Quantisation (GLVQ) model on the training feature matrix, learning one representative prototype per cluster. New clients could be assigned to the closest prototype, enabling real-time segmentation without re-running clustering. The financial and behavioural profiles observed in the test set closely mirrored the training statistics, confirming the stability and representativeness of the learned personas.

In **NB2**, deployment was performed through a simpler mechanism: test clients were assigned to the nearest *K-Means* centroid using the `predict()` function. This provides a zero-cost operational step once clustering has been completed. The difference between training and test Silhouette scores remained below 0.01, per-persona centroid distances were nearly identical across datasets, and cluster proportions in the test set closely matched the training distribution, further supporting the robustness of the segmentation.

7. Bayesian updating framework

Beyond static assignment, both notebooks implement a **Bayesian updating layer** that allows personas to evolve as new client information becomes available, without retraining the clustering model.

Two complementary mechanisms were considered. The **Beta–Binomial update** treats each feature independently: the cluster mean defines a Beta prior, and each new observation shifts the posterior mean toward the observed value while progressively reducing uncertainty. This approach is computationally lightweight and naturally handles partial or asynchronous data availability.

The **Multivariate Gaussian (Kalman) update** instead models the full feature vector jointly using the empirical covariance structure of each training cluster. In this framework, changes in one dimension automatically propagate to correlated features, leading to more coherent and behaviourally consistent persona adjustments when richer information is available.

Sequential updates applied on test-set clients produced small and stable shifts in persona profiles, indicating that the original cluster centroids captured persistent patterns in the broader client population rather than sample-specific artefacts.

Synthesis. NB1 provides a statistically robust segmentation tailored to mixed-type data, while NB2 ensures fast and scalable deployment. Together, they balance methodological rigour and operational efficiency.